TO-DO LIST

REMAS ALIESSA
KADI ALSUBAIE
KADI ALBAKRI

INTRODUCTION:

Our project, “To-Do List Application,” was developed in C++.
It helps users manage tasks by adding, updating, deleting, and marking them as completed.
We used a singly linked list for flexibility and easy updates.

PROJECT IDEA (PROBLEM)

People often forget tasks without an organized system.
Our project provides a simple TO-DO LIST using a SINGLY LINKED LIST to easily add, edit, delete, and complete tasks.

OBJECTIVES

- Build a functional To-Do List application.
- Apply the linked list data structure to manage tasks dynamically.
- Strengthen our skills in C++, classes, pointers, and dynamic memory.
- Use data structures to create a practical, user-focused solution.

CODE EXPLANATION

- We used a singly linked list to store tasks dynamically and flexibly.
- Each task has:
  - description (string)
  - completed (bool)
  - next (pointer to the next task)

CODE EXPLANATION

- We designed a class TodoList with the following methods:
  - Constructor: Initializes head to NULL.
  - Destructor: Frees memory at the end of the program.
  - addTask: Adds a new task to the list.
  - deleteTask: Deletes a task by position.
  - markCompleted: Marks a task as completed.
  - displayTasks: Displays all tasks with their status.
  - updateTask: Updates a task description

RESULT

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===== TO-DO LIST MANAGER =====

Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 1
Enter task description: Getting prepared for data structure assignment
Task added successfully!

Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 1
Enter task description: Submitting our project
Task added successfully!

Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 5

----- YOUR TASKS -----
Task 1: Getting prepared for data structure assignment [Not Completed]
Task 2: Submitting our project [Not Completed]
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Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 4
Enter task position to update: 2
Enter new description: Submitting our TO-DO List project
Task "Submitting our project" updated to "Submitting our TO-DO List project".

Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 3
Enter task position to mark as completed: 1
Task "Getting prepared for data structure assignment" is now completed!

Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 5

----- YOUR TASKS -----
Task 1: Getting prepared for data structure assignment [Completed]
Task 2: Submitting our TO-DO List project [Not Completed]
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Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 2
Enter task position to delete: 1
Task "Getting prepared for data structure assignment" has been deleted.

Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 5

----- YOUR TASKS -----
Task 1: Submitting our TO-DO List project [Not Completed]
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Menu:
1. Add a new task
2. Delete a task
3. Mark task as completed
4. Update a task
5. View all tasks
6. Exit
Enter your choice: 6
Exiting program. Goodbye!

Process returned 0 (0x0)   execution time : 172.022 s
Press any key to continue.
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CONCLUSION

- We applied what we learned in class to build a real project.
- We gained experience with dynamic memory, pointers, and class design in C++.
- We learned how data structures solve real-world problems.
- This project improved our technical skills and boosted our confidence.